2018 EXAMINATIONS

ECON 102 PRINCIPLES OF ECONOMICS **(3 hours)**

Candidates should answer ALL questions from Section A, ONE question from Section B and ONE question from Section C.
Non-programmable calculators are allowed in this exam.

Please turn over

Section A (40 marks). Each question carries equal weight.
Identify the correct answers on the answer sheet provided.

1. If price elasticity of demand for a good is 2, this implies that consumers would

(a) buy twice as much of the good if price falls by 10%.
(b) require a 2% cut in price to raise quantity demanded of the good by 1%.
(c) buy 2% more of the good in response to a 1% cut in price.
(d) require at least a £2 increase in price before showing any response to the price increase.

2. Consumers in Timbuktu like to consume soda and popcorn together; moreover, consumers tend to buy more of both goods when they are wealthier. In Timbuktu, the cross-price elasticity of demand between soda and popcorn is _____, while the income elasticity of demand for popcorn and soda is _____.

(a) positive; negative
(b) positive; positive
(c) negative; negative
(d) negative; positive

3. If a drought destroyed half of the French garlic crop at a time when the health benefits of garlic were being well publicized, economists would expect that in the market for garlic

(a) quantity exchanged would rise but the change in price is uncertain without further information.
(b) price would rise but the change in quantity exchanged is uncertain without further information.
(c) both price and quantity exchanged would rise.
(d) price would rise, and quantity exchanged would fall.

4. In Lancaster, the supply for cheeseburgers is $P = 3 + 4Q$, while demand is $P = 17 – 3Q$, where P is the market price of burgers, and Q is the quantity of burgers in thousands. In equilibrium, the market price is

(a) less than 8.
(b) between 8 and 10.
(c) between 10 and 12.
(d) greater than 12.

5. Continuing with question 4. In equilibrium, consumer surplus is

(a) less than 7,000.
(b) between 7,000 and 9,000.
(c) between 9,000 and 11,000.
(d) greater than 11,000.

Please turn over

6. Roy divides his income between Morrissey CDs and posters of Kim Kardashian. At his current level of consumption, Roy is willing to trade off CDs for posters at the rate of 1 CD for 3 posters. If Roy is maximizing his utility subject to his budget constraint, then

(a) the price of a CD must be 3 times greater than the price of a poster.
(b) the price of a CD must be 1/3 of the price of a poster.
(c) at the optimal bundle, the marginal utility of CDs must be 3.
(d) at the optimal bundle, the marginal utility of CDs must be 1/3.

7. Tim spends his £36 income on goods x and y. The marginal rate of substitution between goods x and y is $MRS_{x,y} = \frac{y}{2x}$. The unit price of good x is 2 and the unit price of good y is 3. The consumption bundle that maximizes Tim's income subject to his budget constraint consists of ____ units of good x and ____ units of good y.

(a) 4; 12
(b) 3; 10
(c) 8; 20/3
(d) None of the above

8. Sheila is searching for a new flat in Lancaster. There are many flats available that rent for the same price, but they differ in quality. Sheila is risk-neutral and values a high-quality flat at £600; she values a low-quality flat at £500. 1/4 of the flats are high quality, and 3/4 are low quality, but Sheila can only determine the quality by visiting. The first flat she visits is low quality. Sheila should visit the second flat as long as the opportunity cost of doing so is

(a) less than £20.
(b) less than £25.
(c) less than £30.
(d) less than £35.

9. Christine divides her income between ramen noodles and filet mignon steak. For Christine, ramen noodles are an inferior good, while steak is a normal good. Following a decrease in the price of ramen noodles, the income effect drives Christine to consume _____ ramen noodles, and the substitution effect leads her to consume _____ ramen noodles.

(a) more; more
(b) more; less
(c) less; more
(d) less; less

Please turn over

10. The manufacturing process for doo-dads releases a significant amount of pollution. If left alone, the market for doo-dads will likely lead to _____ of doo-dads, relative to the socially optimal level. In such circumstances, a _____ could help lead the market towards the socially efficient level of production.

(a) overproduction; tax on production of doo-dads
(b) underproduction; tax on production of doo-dads
(c) overproduction; subsidy for consumption of doo-dads
(d) underproduction; subsidy for production of doo-dads

11. Consider the production function $Y = K^{½}L^{½}$ where Y denotes output, K is capital stock and L is labour input. The 10-unit isoquant passes through the point where

(a) K=4 and L=20
(b) K=5 and L=20
(c) K=3 and L=20
(d) None of the above

12. The average cost of a firm in a perfectly competitive industry is given by $5q^2-2q$ for all values of q greater than zero. In long run equilibrium, the price will be

(a) 100 or less
(b) more than 100 but not more than 110
(c) more than 110 but less than 120
(d) 120 or more

13. A profit-maximising monopolist facing a linear demand curve will produce

(a) on the elastic part of the demand curve
(b) on the inelastic part of the demand curve
(c) on the point of the demand curve where the (absolute) elasticity is one
(d) on the point of the demand curve where the (absolute) elasticity is zero

14. Monopolies impose a welfare loss because

(a) at the profit maximising output level, the firm chooses not to sell to some potential customers who are willing to pay more than the marginal cost of production
(b) at the profit maximising output level, the firm chooses not to sell to some potential customers who are willing to pay more than the average revenue
(c) the gain in consumer surplus is offset by a reduction in producer surplus
(d) they do not produce as much as perfectly competitive firms, so workers become unemployed and receive welfare payments

Please turn over

15. A profit-maximising price-discriminating monopolist selling output to two distinct markets chooses prices in each market so that

(a) the demand curve is the same in both markets
(b) average cost is the same in both markets
(c) average revenue is the same in both markets
(d) marginal revenue is the same in both markets

16. Under monopolistic competition, long run equilibrium occurs where

(a) average revenue is tangent to the marginal cost curve
(b) marginal revenue is tangent to the marginal cost curve
(c) average revenue is tangent to the average cost curve
(d) marginal revenue is tangent to the average cost curve

17. In Cournot-Nash equilibrium

(a) each player recognises that no unique solution exists
(b) each player maximises their own payoff given other players' actions
(c) each player maximises the joint payoff of all players in the game
(d) there must be a prisoners' dilemma

18. Consider the game represented by the following payoff matrix. Within each cell, the first number refers to the payoff earned by Rowan, while the second refers to the payoff earned by Colin. Rowan chooses one of two actions, represented by the rows of the matrix. Colin chooses one of four actions, represented by the columns of the matrix. Which of the following can be said about the solution to this game?

	Colin plays A	Colin plays B	Colin plays C	Colin plays D
Rowan plays X	10,8	10,5	12,4	8,4
Rowan plays Y	9,6	11,4	10,3	6,5

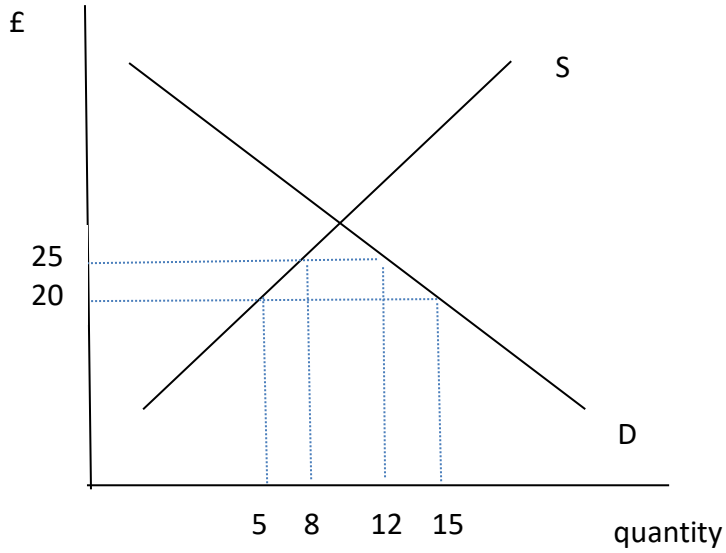
(a) Rowan will play X and Colin will play A
(b) Rowan will play Y and Colin will play D
(c) Rowan will play X and Colin will play B
(d) the game is not dominance solvable

19. The backward bending supply curve of labour occurs when

(a) the income effect of a price increase dominates the substitution effect
(b) the substitution effect of a price increase dominates the income effect
(c) the income effect of a wage increase dominates the substitution effect
(d) the substitution effect of a wage increase dominates the income effect

Please turn over

20. Consider the following diagram which shows the impact of imposing a tariff of £5 on each unit of an internationally traded product that is imported, where the world price is £20. Before the imposition of the tariff, domestic producers supply 5 units to the market and 10 units are imported. After the imposition of the tariff domestic producers supply 8 units to the market and 4 units are imported.



What is the welfare loss due to imposition of this tariff?

(a) 10
(b) 15
(c) 20
(d) none of the above

21. In a certain fictional country the value of Output (in billions) is 1,600$, the population number is 80 million and the number of people employed is 40 million. What is the output per person and the average labour productivity in this country?

(a) Output per person=20,000$; Average labour productivity=40,000$ per person.
(b) Output per person=40,000$; Average labour productivity=20,000$ per person.
(c) Output per person=20$; Average labour productivity=40$ per person.
(d) Output per person=30,000$; Average labour productivity=50,000$ per person.

22. In a leading exporting country, the revenue as a percentage of GDP is equal to 80% whereas the expenditure out of GDP is equal to 30%. What is the value of government deficit or surplus in this country?

(a) Surplus of 50 percent out of GDP.
(b) Surplus of 80 percent out of GDP.
(c) Deficit of 50 percent out of GDP.
(d) Deficit of 110 percent out of GDP.

Please turn over

23. The demand equation for employment is given by $N_d = 600 - 3w$ whereas the supply of labour is given by $N_s = 100 + 2w$. The terms N_d, N_s and w represent labour demand, labour supply and wages, respectively. What is the equilibrium level of employment and wages in this economy?

(a) w=100 and N=300.
(b) w=150 and N=200.
(c) w=100 and N=370.
(d) None of the above.

24. Which of the following is *not* classed as a stabilization policy?

(a) Finance minister cutting the rate of income tax to promote employment.
(b) The U.K negotiating its trade policies with the EU following Brexit.
(c) Central bank lowering interest rates to stimulate the economy.
(d) The government deficit increases as the economy moves into recession.

25. For an economy characterized by: C=1800+0.60(Y-T), I=800, G=1,500, NX=100, T=1,500, what is the equilibrium level of output (Y)?

(a) 8000.
(b) 8250.
(c) 8500.
(d) 8750.

26. If living standards in a country, as measured by output per person, increase, then total output must have:

(a) increased more rapidly than the population increase.
(b) increased at the same rate of the population increase.
(c) increased more slowly than the population increase.
(d) decreased more rapidly than the population decrease.

27. Compound interest is:

(a) the payment of interest on the original deposit.
(b) the interest rate adjusted for the rate of inflation.
(c) the real rate of interest compounded by the rate of inflation.
(d) the payment of interest on the original deposit and all accumulated interests.

28. In the long run, increases in output per person arise primarily from:

(a) increases in male labour force participation.
(b) an increasing proportion of the population retiring.
(c) a decreasing proportion of the population retiring.
(d) increases in average labour productivity.

Please turn over

29. Increasing the capital available to the workforce, holding other factors constant, tends to _____ total output at a(n) _____ rate.

(a) decrease; increasing
(b) increase; decreasing
(c) decrease; decreasing
(d) increase; constant

30. The real rate of interest measures the _____ of capital investment.

(a) equilibrium quantity
(b) opportunity cost
(c) relative price
(d) marginal benefit

31. In the open economy IS-LM model, a simultaneous increase in the money supply and a fall in net exports will:

(a) increase the level of output.
(b) reduce the level of output.
(c) increase the interest rate.
(d) reduce the interest rate.

32. Monetary policy will be more effective when _____.

(a) the IS curve is relatively flat.
(b) the country is in a liquidity trap.
(c) the IS curve is relatively steep.
(d) the opportunity cost of holding money is low.

33. If a government decides to lower its deficit by limiting funds to the National Health Service (NHS), then according to the IS-LM model:

(a) The IS curve shifts to the right.
(b) Investments will be lower.
(c) Interest rates will be lower.
(d) none of the above.

34. If changes in the money supply do not change the rate of interest, we can say there is _____.

(a) a lot of scope for monetary policy.
(b) a recessionary gap.
(c) an expansionary gap.
(d) a liquidity trap.

Please turn over

35. If taxes increase, the IS-LM equilibrium will_____. (Note that *Y* refers to output and *i* to the interest rate).

(a) yield a higher *Y* and *i*.
(b) yield a lower *Y* and *i*.
(c) yield a lower *Y* but a higher *i*.
(d) yield a higher *Y* but a lower *i*.

36. In presenting the expectations-augmented Phillips curve, Milton Friedman *assumed* that

(a) trade union activity is the exogenous variable, that causes variations in unemployment
(b) monetary policy is the exogenous variable, that causes variations in unemployment
(c) unemployment is the exogenous variable, that causes variations in inflation
(d) monetary policy is the exogenous variable, used to counter variations in unemployment

37. The natural rate of unemployment is the level of unemployment that is consistent with

(a) a high rate of inflation
(b) a low rate of inflation
(c) the absence of monetary disturbance
(d) the absence of involuntary unemployment

38. With quantitative easing a central bank

(a) purchases long-term interest-bearing bonds
(b) causes bond prices to rise
(c) increases the volume of base money
(d) all of the above

39. Milton Friedman's monetarism is a theory of

(a) aggregate demand management
(b) the demand to hold money
(c) the money supply
(d) none of the above

40. A repurchase agreement requires a commercial bank to

(a) repurchase bonds, previously sold to the central bank.
(b) resell bonds, previously bought from the central bank.
(c) extend loans to customers, against collateral purchases
(d) extend loans to customers, against collateral sales

Please turn over

Section B (30 marks) Answer ONE question

41. (a) Briefly explain what is a public good and give 3 examples of such goods. Explain why public goods are often provisioned by government entities and not the market. (10 marks)

(b) Explain the Coase Theorem, and provide a numerical example to help illustrate the concept. (10 marks)

(c) Explain the relevance of the assumption of "zero transactions costs" for the Coase Theorem. Give an example of where this assumption might fail in practice, and how government intervention could help rectify the situation. (10 marks)

42. Using an appropriate diagram, explain how a minimum wage can lead to an increase in employment under conditions of labour market monopsony. What impact does the minimum wage policy have on economic welfare? (30 marks)

43. (a) Determine the values of x and z that maximise the function
$y = 10x - \frac{5}{2}x^2 + 2xz - 2z^2$ (15 marks)

(b) A firm can sell its products in two countries. Quantity demanded in the first country is given by

$q = \frac{99}{2} - \frac{1}{2}p$

and in the second country

$$Q = 99 - P$$

The firm's total output is q+Q which it can produce at a total cost (TC) of

$$TC = 50q + 50Q + qQ + 0.5q^2 + 0.5Q^2$$

Find
(i) the prices and outputs in each country that maximise profits (9 marks)
(ii) the price elasticity of demand in each country at the profit maximising prices (6 marks)

Please turn over

Section C (30 marks) Answer ONE question

44. Consider the following economy:

Consider the following economy:

$$C = 200 + 0.75(Y - T)$$

$$I = 200 - 25r$$

$$T = 100, G = 100$$

$$\left(\frac{M}{P}\right)^d = Y - 100r$$

$$P = 2. M^s = 1000$$

C=consumption, Y=output, I=investment, T=taxes, G=government spending, r=interest rate, $\left(\frac{M}{P}\right)^d$=real demand for money, P=prices and M^s=money supply.

a. Find the algebraic expression for the IS curve and the LM curve and explain why the slope of the IS curve is negative while the slope of the LM curve is positive. (10 marks)

b. Find the equilibrium interest rate (r) and the equilibrium level of output (Y). (10 marks)

c. Suppose the government increases its purchases from 100 to 150. What is the new equilibrium level of interest rate and output? Explain your answer using words and a relevant graph. (10 marks)

45. Explain the concept of a liquidity trap and its implications for monetary policy? Which type of policies can be effective in releasing the economy from a liquidity trap? Illustrate and explain using the IS-LM model. (30 marks)

46. Explain the concept of the zero lower bound to short-term interest rates, and the relevance of quantitative easing to that situation. (30 marks)

End of paper